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Synthesis of ultrathin WSe₂ nanosheets and their high-performance catalysis for conversion of amines to imines†

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Tungsten diselenide (WSe₂) is the material with the lowest thermal conductivity in the world. Most physical methods are used for the synthesis of tungsten diselenide. Here, a simple colloidal method is reported for the synthesis of WSe₂ nanosheets. The composition, valence, size, morphology and properties of the samples were characterized and measured. Results showed that the obtained WSe₂ nanosheets with a thickness of 0.7 nm had strong blue fluorescence. Significantly, the synthesized WSe₂ nanosheets exhibited excellent catalytic activity for the aerobic coupling of amines to imines, with 100% yield under visible light irradiation and air atmosphere. As a photocatalyst, it exhibited excellent recyclability, and maintained a high yield after 5 cycles. It was found that this reaction could also happen in the presence of natural light by slightly extending the reaction time. Moreover, H₂O was used as a solvent in the catalytic process, avoiding expensive and toxic organic solvents. This work provides an efficient, economical and sustainable process for the synthesis of imines and shows the great potential of WSe₂ nanosheets as photocatalysts for organic synthesis.

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Introduction

Tungsten diselenide (WSe₂), an important transition metal dichalcogenide, has received extensive attention because of its typical layered crystal structure.^{1–3} At present, the performance of tungsten diselenide has mainly been focused on the fields of diodes,⁴ batteries,⁵ field effect transistors^{6,7} and hydrogen evolution reaction (HER).^{8,9} The two-dimensional (2D) layered materials possess larger surface areas and more active sites than their bulk counterparts, helping to improve the performance of the material. Therefore, 2D materials with hierarchical architectures attract more and more attention. Zhao *et al.* prepared high quality WSe₂ nanocrystals using a chemical vapor transport method.¹⁰ Browning *et al.* successfully synthesized uniform few-layer WSe₂ with an atomically smooth surface through a selenium–oxygen ion exchange process.¹¹ Zou *et al.* designed and synthesized 3D dendritic WSe₂ on a conductive carbon nanofiber mat (d-WSe₂/CFM) by a diffusion-controlled chemical vapor deposition method.¹² Tan *et al.* exfoliated

WSe₂ from bulk using aromatic intercalants, such as phenyllithium, sodium naphthalenide, and sodium anthracenide.¹³ However, these above-mentioned methods are excessively demanding on the production equipment and conditions, which is not conducive to mass production. There have been very few reports of solution-phase syntheses of colloidal WSe₂ nanostructures. Antunez *et al.* synthesized nano-onion structured WSe₂ nanosheets by the injection of di-*tert*-butyl diselenide (*t*-Bu₂Se₂) into a solution of WCl₄ at 225 °C for several hours.¹⁴ Jin *et al.* synthesized single-layer WSe₂ quantum dots by a mixture of diphenyl diselenide (Ph₂Se₂) and TOPO injected into a solution of tungsten hexacarbonyl (W(CO)₆) at 330 °C for 1 hour.¹⁵ Therefore, it is urgently needed to propose a simple method for synthesizing WSe₂ nanosheets.

Imines and their derivatives are important intermediates of a series of fine chemicals, dyes, pharmaceuticals, agrochemicals, polymers, lipoxygenase inhibitors, and anti-inflammatory and anti-cancer agents.^{16–19} The oxidation of amines to the corresponding imines is an efficient way to obtain imines.^{20,21} Traditional thermochemical methods usually require high temperature conditions, while photochemical methods for synthesizing imines can generally be performed in a mild environment. Natural light is a sustainable developmental force for amine oxidative coupling reactions. Those photocatalysts containing precious metals such as Pt,²² Pd,^{23,24} and Au^{25,26} are unfavorable for large-scale applications because of

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their low abundance and high cost. For this reason, it is crucial to develop non-precious metal photocatalysts for the synthesis of imines.

In the current work, we synthesized ultrathin WSe₂ nanosheets by the colloidal method. The synthesized WSe₂ nanosheets showed excellent catalytic performance and high reusability for coupling of various amines to corresponding imines. We propose a green, economical and sustainable synthesis method of imines using eco-friendly oxygen as an oxidant from the air and abundant water as a solvent rather than a complex toxic organic compound. The reaction mechanism for oxidative coupling of amines to imines is also speculated. To the best of our knowledge, there is no report on the use of tungsten diselenide as a photocatalyst in organic synthesis reactions.

Experimental section

Chemicals and characterization (see the ESI†)

Synthesis of WSe₂ nanosheets. The WSe₂ nanosheets were synthesized using a vacuum/nitrogen gas Schlenk line. Typical synthesis method: Firstly, for the preparation of selenium precursor solution (A), 2.4 mmol selenium powder (Se), 2.4 ml oleylamine (OLA) and 2.4 ml 1-dodecanethiol (1-DDT) were placed in a 50 ml three-necked flask in turn, and after the flask was connected to the reaction system as shown in Fig. S1,† vacuum and nitrogen was circulated three times to remove oxygen from the reaction equipment; then, the mixture was dissolved with magnetic stirring under a nitrogen atmosphere. Secondly, for the preparation of cation solutions (B), 1 mmol tungsten hexachloride (WCl₆), 10 ml OLA, 10 ml oleic acid (OA) and 10 ml 1-octadecene (ODE) were mixed in a 100 ml three-necked flask undergone cycles between vacuum and nitrogen in turn for removing the oxygen of reaction system; then, the mixture solution was raised to 140 °C and evacuated for 30 minutes to remove the low boiling point substances. Finally, solution (B) was heated to 300 °C under a nitrogen atmosphere, and then the selenium precursor solution (A) was injected into solution (B), and the reaction was maintained for one hour under magnetic stirring (Fig. S2†). The product was collected by centrifugation and washed using chloroform and ethanol several times, followed by vacuum drying at 60 °C for further characterization and measurement.

Catalysis of amine coupling reaction

0.5 mmol amine and 30 mg of photocatalysts (WSe₂) were dispersed in 0.5 ml of solvent in a 10 ml Schlenk reactor. The coupling reaction was carried out in an oil bath with magnetic stirring and irradiated under a 5 W white LED lamp. After the completion of the reaction, the catalyst was removed by centrifugation. Then, the reaction mixture was extracted with ethyl acetate (2 mL × 3). The final solution was analyzed using gas chromatography (GC) and gas chromatography-mass spectrometry (GC-MS).

Results and discussion

Fig. 1a shows the X-ray diffractometer (XRD) patterns of the obtained samples, and the diffraction peaks at 13.62°, 32.17°, 37.80°, 47.36° and 66.22° match well with the (002), (100), (103), (106) and (008) crystal planes of the hexagonal WSe₂ system with a space group *P*6₃/*mmc*, respectively (JCPDS card no. 38-1388). The transmission electron microscope (TEM) images shown in Fig. 1b and c demonstrate that the WSe₂ nanosheets were composed of flowerlike nanostructures including many ultrathin nanosheets. The inset in Fig. 1c shows the selected area electron diffraction (SAED) patterns of the ultrathin nanosheets, revealing that the *d*-spacing values of 0.652, 0.287, 0.216, 0.172 and 0.162 nm correspond to the (002), (100), (103), (106) and (008) planes of WSe₂, respectively. Fig. 1d shows the high resolution transmission electron microscopy (HRTEM) image and that the lattice fringes of 0.652 nm and 0.287 nm correspond to the (002) and (100) planes of WSe₂. The atomic force microscope (AFM) image and the corresponding height profiles are shown in Fig. 1e and f. The average height of the edge of the WSe₂ nanosheet is about 0.7 nm, which is consistent with the height of the single-layered WSe₂ nanosheet shown in previous reports,^{27,28} indicating that a part of the monolayer nanosheet is present in the obtained product.

The chemical compositions of the samples are determined using X-ray photoelectron spectroscopy (XPS). As shown in

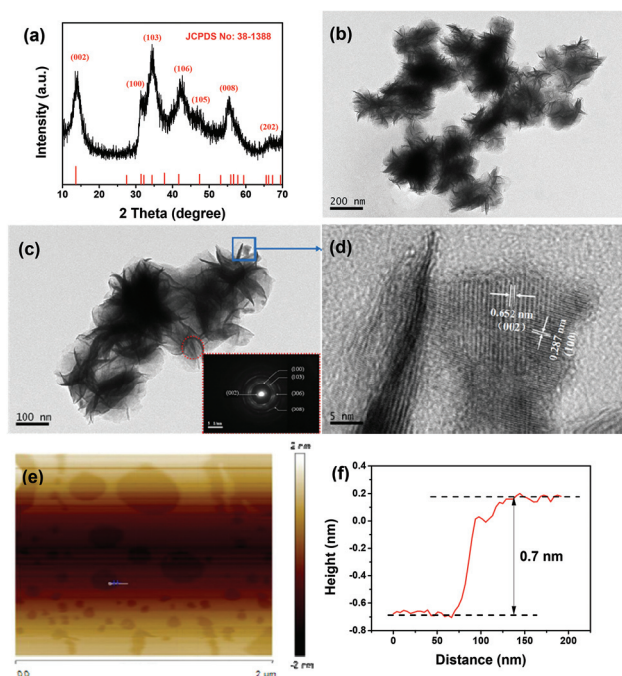


Fig. 1 (a) XRD patterns of the obtained samples and the standard XRD patterns of WSe₂; (b), (c) TEM images of WSe₂ nanosheets and the inset shows the corresponding SAED pattern; (d) HRTEM image of the selected region; (e) AFM image of WSe₂; and (f) height profiles from sections as indicated by the white lines of WSe₂.

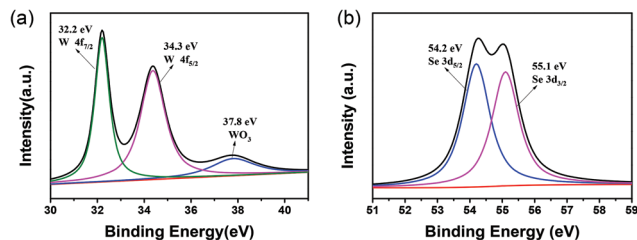


Fig. 2 XPS analysis of W 4f and Se 2p peaks of WSe₂, respectively.

Fig. 2a, two characteristic peaks at around 32.2 and 34.3 eV are attributed to the 4f_{7/2} and 4f_{5/2} for WSe₂.¹² Additional weak peaks of W 4f_{3/2} located at around 37.8 eV are assigned to WO₃ due to oxidation of surface W under air exposure conditions,^{29–31} and the W 4f_{7/2} from WO₃ at 35.7 eV is buried by the strong peak of W 4f_{5/2} from WSe₂.³² In Fig. 2b, we can see the binding energies of Se 3d_{5/2} and Se 3d_{3/2} at 54.2 and 55.1 eV, which are attributed to divalent Se ions.^{33–35} The XPS results further illustrate that the valence of the obtained samples is in conformity with the characterization of the XRD pattern.

Fig. 3a shows the UV-Vis absorption and Tauc plot of WSe₂ nanosheets. There is a broad absorption in the range of 200 to 800 nm. Then the band gap is followed according to the Tauc formula: $(\alpha h\nu)^2 = C'(h\nu - E_g)$, where α , $h\nu$ and C' are the absorption coefficient, photoenergy and a constant, respectively.^{36,37} The band gap energy of WSe₂ nanosheets obtained by extrapolating the linear part of the curves of $(\alpha h\nu)^2$ versus $h\nu$ has been deduced to be 1.7 eV compared with the band gap energy of the bulk WSe₂ at 1.2 eV.³⁸ The transition from the indirect band gap in bulk WSe₂ to the direct band gap in the monolayer WSe₂ can emit photoluminescence (PL).

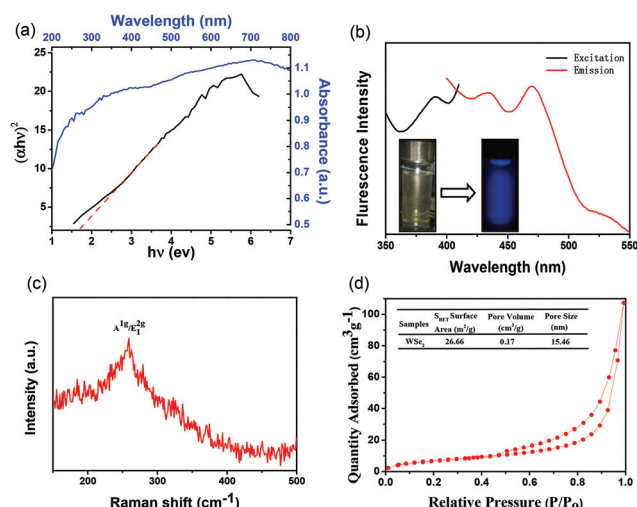


Fig. 3 (a) UV-Vis absorption spectra of WSe₂ (black) and Tauc plot (blue) of WSe₂; (b) photoluminescence excitation (PLE) spectra of WSe₂ and the inset shows the WSe₂ nanosheets dispersed in NMP under visible light and UV light of 365 nm, respectively; (c) Raman spectrum of WSe₂; and (d) the N₂ adsorption-desorption isotherm of the WSe₂.

Photoluminescence excitation (PLE) spectra of WSe₂ powder showed a maximum excitation wavelength at 390 nm, a maximum emission wavelength at 470 nm, and a shoulder peak at about 433 nm (Fig. 3b). The inset in Fig. 3b exhibits the color change of WSe₂ nanosheets dispersed in *N*-methyl-2-pyrrolidinone (NMP) from faint yellow to conspicuous blue under UV light irradiation at 365 nm. As for the Raman spectrum of WSe₂ nanosheets in Fig. 3c, only one peak was observed from 200 to 300 cm⁻¹ due to the small energy difference between A_{1g} and E_{2g}¹ modes, at 253 and 250 cm⁻¹, respectively.^{39,40} Fig. 3d displays the N₂ adsorption-desorption isotherm of the WSe₂ nanosheets, and the *S*_{BET} surface area is 26.66 m² g⁻¹, with a pore volume of 0.17 cm³ g⁻¹ and a pore size of 15.46 nm.

The photocatalytic activity of the WSe₂ nanosheets was determined in the oxidative coupling reaction. As shown in Table 1, the model substrate of 0.5 mmol benzylamine was reacted with 30 mg WSe₂ under irradiation using a 5 W white LED light at 60 °C. With 1 atm of air as the oxidant and 0.5 ml H₂O as the solvent, benzylamine was converted into *N*-benzylidene benzylamine with a high yield of 100% (Table 1, entry 1), which is better than most reported oxidative coupling reactions summarized in Table S1.[†]^{19,36,37,42,53–58} The high performance in the aerobic coupling of amines to imines could be due to the typical layered crystal structure of tungsten diselenide (WSe₂) with a large surface area and more active sites than their bulk counterparts, helping to improve the catalytic performance of the material. Moreover, it is worth noting that the obtained WSe₂ nanosheets exhibited strong and broad absorption in the visible region, which is active for visible-light photocatalytic reactions. In addition, almost no desired product was obtained without the addition of WSe₂ nanosheets (Table 1, entry 2). The reaction poorly occurred in a dark box (Table 1, entry 3), which confirms that the oxidative

Table 1 Oxidative coupling of benzylamine by WSe₂ nanosheets

Entry	WSe ₂	Solvent	Time (h)	Yield (%)
1	✓	H ₂ O (0.5 ml)	24 h	100
2	×	H ₂ O (0.5 ml)	24 h	Trace
3 ^a	✓	H ₂ O (0.5 ml)	24 h	19
4 ^b	✓	H ₂ O (0.5 ml)	30 h	98
5	✓	H ₂ O (1.0 ml)	24 h	89
6	✓	H ₂ O (5.0 ml)	24 h	36
7	✓	Acetonitrile (0.5 ml)	24 h	96
8	✓	Cyclohexane (0.5 ml)	24 h	95
9	✓	<i>N</i> -Hexane (0.5 ml)	24 h	83
10	✓	Ethanol (0.5 ml)	24 h	70
11	✓	<i>N,N</i> -Dimethylformamide (0.5 ml)	24 h	63
12 ^c	✓	H ₂ O (0.5 ml)	24 h	Trace
13 ^d	✓	H ₂ O (0.5 ml)	24 h	7

Reaction conditions: 0.5 mmol amine, 30 mg WSe₂, air, a 5 W white LED lamp, 60 °C, determined using GC. ^a Response in the dark cabinets. ^b The reaction takes place under natural light with no extra light source. ^c The reaction was performed under a nitrogen atmosphere. ^d 0.5 mmol amount of *p*-benzoquinone was added as a superoxide scavenger.

coupling reaction of benzylamine is a light-driven reaction. In particular, the conversion up to 98% can be achieved relying on natural light from indoors without the need for additional light sources by prolonging the reaction time up to 30 h (Table 1, entry 4). These results clearly show that the oxidative coupling reaction of benzylamine is catalyzed by WSe₂ nanosheets under very mild visible light conditions. As an additional control experiment of different amounts of solvents, the yields varied widely, with 5 ml, 1 ml, and 0.5 ml of H₂O corresponding to 38%, 89%, and 100%, respectively (Table 1, entries 5, 6 and 1). The reason could be that less solvent increases the effective number of collisions between the catalyst and the reactants. Using WSe₂ nanosheets as the photocatalysts, different solvents were investigated (Table 1, entries 1 and 7–11). In acetonitrile, *N*-hexane, cyclohexane, ethanol and *N,N*-dimethylformamide, the yields were 96%, 95%, 83%, 70% and 63%, respectively. Meanwhile, the photocatalysis under solvent-free reaction for the conversion from amines to imines also follows the same tendency as in several kinds of solvents. As shown in Table S2,† the yield of *N*-benzylidene benzylamine can reach 1.28 mmol, which is close to that of the previously reported work of Yang *et al.* (yield = 1.30 mmol) with mesoporous Au/TiO₂ nanospheres as the catalyst in a solvent-free system and under an air atmosphere.⁴¹ All in all, environmentally friendly H₂O has become the preferred solvent in our catalytic process, and can be also used for all the subsequent oxidative coupling reactions.

The catalytic performance of WSe₂ nanosheets for several amines with electron-withdrawing or electron-donating groups was investigated in order to explore the generality of the catalytic system and the results are collected in Table 2. All reactants are efficiently converted to the corresponding imines, but the conversion efficiency is slightly reduced compared to that of benzylamine (Table 2, entries 1–6). It appears that the

efficiency of the oxidative coupling reaction is dependent on the steric structure (Table 2, entries 4 and 6). In addition, benzylamines with electron-withdrawing groups exhibited higher activity than the substrates having electron-donating groups (Table 2, entries 2–6). This may be due to the presence of methyl groups forming stable cationic radicals that slow down the formation of the final product.⁴²

Further experiments were carried out to research the mechanism of the benzylamine coupling reaction by WSe₂ nanosheets. If the reaction was carried out under a nitrogen atmosphere without oxygen being involved, benzylamine was hardly converted into the product (Table 1, entry 12). This indicates that the oxygen molecules are an important part of the coupling reaction. Previous reports have indicated that molecular oxygen acquires excited electrons from the conduction band of the photocatalyst and then converts into superoxide radicals.^{43,44} In order to confirm that superoxide radicals are involved in the oxidative coupling reaction of benzylamine, *p*-benzoquinone as a superoxide scavenger is added to the reaction mixture (Table 1, entry 13).⁴⁵ In the presence of the superoxide scavenger, the yield of the oxidative coupling reaction was significantly reduced to 7%, indicating that the presence of superoxide radicals is essential for amine coupling reactions in the catalytic reaction.

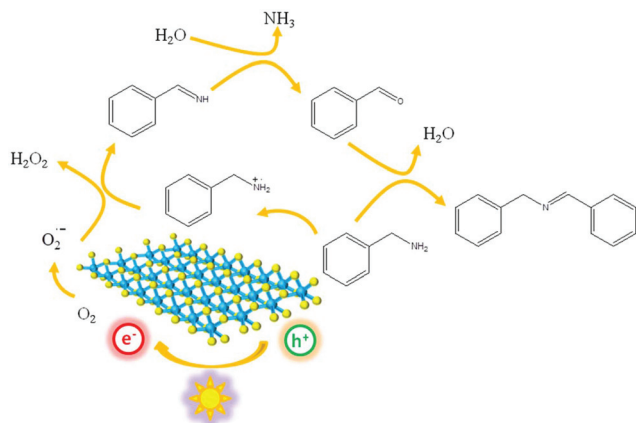
Subsequently, we determined the presence of H₂O₂ in the benzylamine coupling reaction product by an iodimetry experiment using starch as an indicator.^{41,46} As shown in Fig. S4,† the color of the reaction mixture turned to blue, while the color of the benzylamine solution did not change, indicating that iodine (I₂) was formed in the reaction mixture. These results confirmed the presence of H₂O₂ in the reaction mixture, thereby inferring the presence of an oxidation–dehydrogenation process during the benzylamine coupling reaction. Further, we identified the presence of ammonia in the course of the benzylamine coupling reaction mixture. Firstly, the aqueous phase from the reaction solution undergoing extraction of the product with ethyl acetate was placed in a test tube, and then a glass rod moistened with concentrated HCl was placed on top of the solution but not touching the liquid level; white fumes of ammonium chloride were formed, as shown in Fig. S5.†⁴⁷

Based on the above experimental results, we come up with possible routes for the conversion of benzylamine to *N*-benzylbenzylamine as shown in Scheme 1 and the pathways presented below as eqn (a)–(d). Under visible light irradiation, electron–hole pairs are generated in the surface of WSe₂ nanosheets, and O₂ is converted to superoxide O₂^{•−} radicals through electron transfer, while the benzylamine is oxidized to form benzylammonium cations by the holes. Firstly, hydrogen detached from the benzylammonium cation radical by the O₂^{•−} radical forms the intermediate, including benzylimine and H₂O₂; secondly, newly produced benzylimine rapidly reacts with water to form phenylacetaldehyde and NH₃, and phenylacetaldehyde is observed as an intermediate during the oxidative coupling reaction *via* the GC-MS characterization (Fig. S6†). Finally, *N*-benzylbenzylamine is obtained by a con-

Table 2 Oxidative coupling of several amines to imines over WSe₂ nanosheets

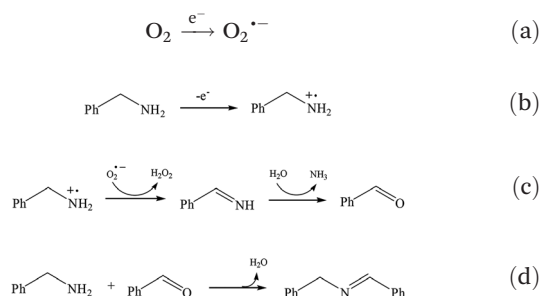
Entry	Reactants	Products	T (h)	Yield (%)
1			24	100
2			24	79
3			24	83
4			24	90
5			24	84
6			24	81

Reaction conditions: 0.5 mmol amine, 30 mg WSe₂, air, a 5 W white LED lamp, 60 °C, determined using GC-MS.



Scheme 1 Proposed mechanism of catalytic reaction for amines to imines in the presence of WSe₂ nanosheets.

condensation reaction between phenylacetaldehyde and another benzylamine.^{48–52}



For further research, the recyclability and stability of WSe₂ nanosheets in the oxidative coupling reaction of benzylamine to imine have been investigated using cycling experiments. After each round of reaction, the WSe₂ nanosheets were separated by centrifugation, washed with ethanol and dried for the next round of reaction. The recovered catalyst retains its excellent yield in the benzylamine coupling reaction even after 5 runs, indicating the great recycling capability of the material (Fig. 4). We also analyzed the crystal phase and morphology of the WSe₂ nanosheets with the XRD pattern and TEM image of the photocatalyst after five recycles (see Fig. S7 and S8†). The XRD pattern and TEM image of the WSe₂ nanosheets remained almost the same as those of the fresh sample.

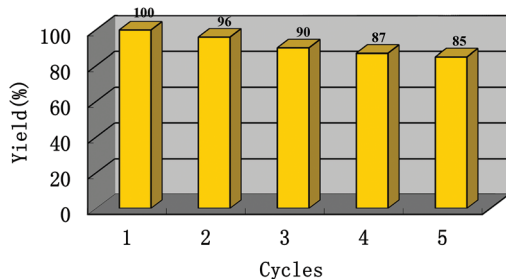


Fig. 4 Recyclability of WSe₂ nanosheets as photocatalysts.

Conclusion

In conclusion, ultrathin WSe₂ nanosheets of 0.7 nm are successfully synthesized by a colloidal method. The obtained nanosheets as photocatalysts exhibited excellent catalytic performance and recyclability for oxidative coupling of benzylamines. Environmentally benign H₂O is used as a reaction solvent. The WSe₂ nanosheets can be considered as low cost, stable and reusable green photocatalysts, which can be expanded to reactions driven by natural light. For all we know, this is the first report on the use of WSe₂ nanosheets as photocatalysts in organic reactions. This work reveals the great potential of WSe₂ nanosheets in the application of organic synthesis.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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